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Schrödinger's Bat: Diffusion Models Sometimes Generate Polysemous Words in Superposition

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Abstract

Recent work has shown that despite their impressive capabilities, text-to-image diffusion models such as DALL·E 2 (Ramesh et al., 2022) can display strange behaviours when a prompt contains a word with multiple possible meanings, often generating images containing both senses of the word (Rassin et al., 2022). In this work we seek to put forward a possible explanation of this phenomenon. Using the similar Stable Diffusion model (Rombach et al., 2022), we first show that when given an input that is the sum of encodings of two distinct words, the model can produce an image containing both concepts represented in the sum. We then demonstrate that the CLIP encoder used to encode prompts (Radford et al., 2021) encodes polysemous words as a superposition of meanings, and that using linear algebraic techniques we can edit these representations to influence the senses represented in the generated images. Combining these two findings, we suggest that the homonym duplication phenomenon described by Rassin et al. (2022) is caused by diffusion models producing images representing both of the meanings that are present in superposition in the encoding of a polysemous word.

https://github.com/rycolab/diffusion-polysemy

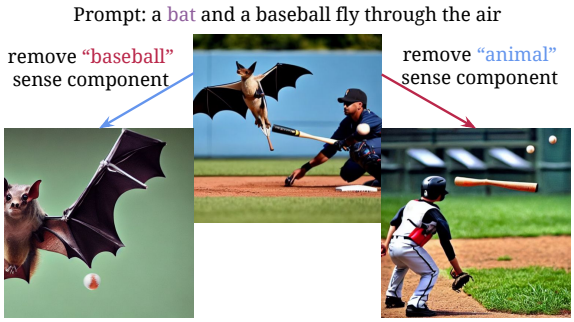

Figure 1: By editing prompt encodings by removing projections of vectors corresponding to each sense, we are able to reduce homonym duplication and produce the desired sense of the word "*bat*".

1 Introduction

When humans encounter a word with more than one possible meaning, they can use information such as situational context and selectional restrictions to infer which meaning is the most likely. Models for language-based tasks such as translation struggle with this, often being biased towards the sense that is more commonly attested regardless of context and thus often producing a translation of the incorrect sense (Campolungo et al., 2022). Recent research by Rassin et al. (2022) showed that text-to-image diffusion models behave in a different and unexpected way when confronted with a prompt containing polysemous words. Specifically, they showed that when DALL·E 2 is prompted with a sentence containing a word with multiple possible meanings, such as "*bat*" or "*bow*", the generated image often represents *more than one* of these meanings. For example, in the initial image in Fig. 1 "*bat*" is represented both as a baseball bat and an animal. They call this behaviour homonym duplication. Diffusion models have attracted much attention, even beyond the reaches of the research communities that spawned them.[1] Despite this, we are still lacking in our understanding of some of the unusual tendencies they exhibit, such as the phenomenon of homonym duplication.

In this work, we put forward one possible explanation of this phenomenon. We suggest that this occurs due to polysemous words in the prompt being encoded as a linear superposition of their possible meanings. In a text-based decoding task, such as translation, a representation of this form would not pose a problem, as the decoding model ultimately converts its representations into a prob-

[1] For example, the Twitter account https://twitter.com/weirddalle, which posts images generated by diffusion models, has over 1 million followers at time of writing.

ability distribution over tokens which it uses to select an output word. Thus, although both possibilities will influence the distribution obtained, it is forced to select a word representing one meaning or the other. However, diffusion models use the representation as input to condition the denoising process which predicts noise to remove, and such a choice between meanings is never forced. This means that both meanings influence the denoising process, and both meanings can be represented in the final image. To draw an analogy with quantum mechanics, in a famous thought experiment, Schrödinger (1935) described a scenario in which a cat in a box with a radioactive material is poisoned once a particle decay occurs, and suggested that since a particle's state exists in a superposition of decayed and undecayed states until an observation is made, that the cat must be considered to be both alive and dead until the box is opened. In the translation scenario, the meaning of a word such as "*bat*" exists as a superposition of possibilities until an "observation" is made by sampling an output word. Since this never happens in diffusion models, the metaphorical box is never opened and the bat continues to be *both* a baseball bat and an animal.

Unfortunately, the internal workings of DALL·E 2 are not openly accessible, so we test this hypothesis by performing experiments using the implementation of Stable Diffusion (Rombach et al., 2022) available on HuggingFace (Wolf et al., 2020). This is a smaller diffusion model which encodes prompts using a CLIP (Radford et al., 2021) encoder, the same type of encoder used by DALL·E 2. We begin by showing that diffusion models can represent multiple concepts that are in linear superposition, by demonstrating that when Stable Diffusion takes as input a sum of CLIP encodings for two words, both objects can be represented in the generated image (§3). We then demonstrate that CLIP encodes polysemous words as a sum over their possible meanings, and that linear-algebraic manipulation of these representations can be used to generate one sense over another (§4), as demonstrated in Fig. 1. Combined, these two facts offer a possible explanation for the homonym duplication described by Rassin et al. (2022).

2 Background: Diffusion Models

Text-to-image diffusion models such as DALL·E 2 (Ramesh et al., 2022) and Stable Diffusion (Rombach et al., 2022) take a text prompt as input and generate an image corresponding to the prompt. During training, an image $\mathbf{x}_0$ is transformed into random noise through the progressive addition of noise sampled from a Gaussian distribution whose mean depends on the image's current value, i.e. $\mathbf{x}_t = \mathbf{x}_{t-1} + \boldsymbol{\epsilon}_t$, $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$ where α_t is a value controlling the magnitude of the noise added at each step. The model is then trained to gradually denoise these images, step by step, conditioned on a descriptive caption of the image. Once trained, the model will apply this process to random noise to produce an image consistent with a given prompt.

2.1 CLIP Encoding

Both DALL·E 2 and Stable Diffusion encode text prompts using CLIP (Radford et al., 2021). CLIP is trained by jointly training a transformer-based text encoder and a transformer-based image encoder on pairs of images and captions with a contrastive objective – given N pairs, it is trained to maximise the cosine similarity between the text and image encodings for these N pairs, but also to minimise cosine similarity between the $N^2 - N$ other pairings.

2.2 Conditioned Denoising Process

Given a prompt $\mathbf{s}$, diffusion models sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ as their initial image. This image is progressively denoised by taking $\mathbf{x}_{t-1} = \mathbf{x}_t - A_t\widehat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t \mid \mathbf{s}) + B_t$, where A_t and B_t are coefficients calculated by a scheduler. To allow generation to be guided by the prompt, $\widehat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t \mid \mathbf{s}) = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t \mid \emptyset) + \gamma \cdot (\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t \mid \mathbf{s}) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t \mid \emptyset))$, where $\gamma \geq 1$ describes the strength of guidance and $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t \mid \mathbf{s})$ is the model that has been trained to predict the noise to remove at each step. $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t \mid \mathbf{s})$ is generally implemented with a version of U-Net (Ronneberger et al., 2015).

DALL·E 2. In DALL·E 2, the prompt $\mathbf{s}$ is encoded using CLIP, a CLIP image encoding is predicted using a transformer-based diffusion prior model conditioned on $\mathbf{s}$ and $\mathrm{CLIP}(\mathbf{s})$. The main diffusion model is then conditioned on this image encoding by projecting it to the appropriate dimension and then concatenating it to the attention context at each layer.

Stable Diffusion. In Stable Diffusion, the denoising process takes place entirely in the latent space. The starting point $\mathbf{z}_T$ is sampled and then de-noised conditioned on $\mathrm{CLIP}(\mathbf{s})$. The conditioning is implemented through cross-attention

between U-Net representations and a projection of $\text{CLIP}(\mathbf{s})$. The final output $\mathbf{z}_0$ is decoded to an image using the decoder of a Variational Auto-encoder (Kingma and Welling, 2014).

2.3 Homonym Duplication in DALL·E 2

Rassin et al. (2022) demonstrate a number of strange behaviours exhibited by DALL·E 2. In this work, we are primarily concerned with the phenomenon of homonym duplication – when multiple senses of a polysemous word are realised in one image. They use a number of prompts to show that when a prompt contains a polysemous word such as "*bat*" or "*bow*", rather than realising the sense that the model deems to be most likely, it often realises both senses in the final image. They also point out that this is observed less frequently in smaller models such as DALL·E Mini and Stable Diffusion.

2.3.1 Stable Diffusion and DALL·E 2

In this work we consider Stable Diffusion in lieu of DALL·E 2. We make this choice since our method directly manipulates the representations used in the generation process, which is not possible in publicly available releases of DALL·E 2. Both are similar in that they both text-to-image generation models based on diffusion and a CLIP encoder. They differ primarily in the fact that Stable Diffusion performs the denoising process entirely in the latent space. DALL·E 2 is also a much larger model in terms of the number of parameters. Our hypothesis, if true, would be more difficult to show in a smaller model with less representational capacity. Because of this, we consider it likely that positive conclusions drawn from this work would also apply to DALL·E 2, though we invite replication of our approach using DALL·E 2.

2.3.2 Homonym Duplication in Stable Diffusion

As Rassin et al. (2022) point out, the homonym duplication phenomenon that they observe in DALL·E 2 is seen less frequently in Stable Diffusion. Nonetheless, we were successful in replicating the behaviour for some prompts. Some of these images are shown in Fig. 1 and Fig. 2. More are included in App. A.1.

3 Claim 1: Diffusion can realise multiple concepts in superposition

To construct an explanation for the phenomenon of homonym duplication, we first show that an encoding that is composed of a sum of encodings of multiple words can produce an image in which each word in the sum is represented. When using the model in the standard way, the prompt $\mathbf{s}$ is encoded using CLIP, and the result $\text{CLIP}(\mathbf{s})$ is then used to condition the denoising process. In these experiments, we instead take two prompts $\mathbf{s}_1$ and $\mathbf{s}_2$, and the denoising process is conditioned on the weighted sum of their CLIP encodings $\alpha_1 \text{CLIP}(\mathbf{s}_1) + \alpha_2 \text{CLIP}(\mathbf{s}_2)$, where $\alpha_1, \alpha_2 \in \mathbb{R}$ and $\alpha_1 + \alpha_2 = 1$.[2]

(a) Prompt: "*tall cranes by the sea*"

(b) Prompt: "*a man holding glasses*"

(c) Prompt: "*a gentleman with a bow and arrow*"

(d) Prompt: "*a baseball bat inside a spooky cave*"

Figure 2: Examples of homonym generation observed in images generated using Stable Diffusion

3.1 Experimental Results

For each pair of prompts, $\mathbf{s}_1$ and $\mathbf{s}_2$, 30 images were generated from $\alpha_1 \text{CLIP}(\mathbf{s}_1) + \alpha_2 \text{CLIP}(\mathbf{s}_2)$, 30 images were generated from $\text{CLIP}(\mathbf{s}_1)$ and 30 images were generated from $\text{CLIP}(\mathbf{s}_2)$. Results presented here are for pairs that were observed to result in realisation of both prompts. Some of the images generated using this method are shown in Fig. 3, and table 1 shows the percentage of images generated that contained the object described by $\mathbf{s}_1$, that described by $\mathbf{s}_2$, both or neither. We can see that the model can, in fact, generate an image representing both of the words in the sum. In fact, we see that this occurs in over 40% of generated images. More generated images are included in App. A.2.1 and a detailed breakdown

[2] In practice, we achieved best results with $\alpha_1 = \alpha_2 = 0.5$.